With Subjects, Italian Relative Clauses Attach Low

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Introduction. How language users resolve the ambiguity that arises when a Relative Clause (RC) follows a complex noun phrase (as in 1) is known to vary across individuals and languages. Crucially, we have yet to disentangle the complex array of factors influencing a preference for the RC modifying the first noun (HA) or the second (LA). Consider Italian speakers, who are commonly argued to prefer HA over LA. Past studies providing evidence for this preference have mostly tested RCs modifying nouns in object position. However, Hemforth et al. (2015) argue that in such position the entire complex NP tends to receive broad focus, at thus more naturally attracts HA modification. They predict that having complex NPs in subject position might lead to a lower preference for HA. On the other hand, Grillo & Costa (2014) show an effect of main clause verb type on attachment, related to the availability in Italian of an underlying structure (pseudo-relatives; PR) licensed by perceptual verbs and only compatible with HA interpretations (cf. Lee & De Santo 2024, De Pedis et al. 2025). Together, these observations complicate the view of Italian as an "HA language". Here, we explore whether **a)** attachment preferences in Italian are affected by **the Position** of the complex-NP + RC and **b)** we can replicate the offline **Verb Type** effect of Grillo & Costa (2014).

Experiment 1: Materials. Target items were globally ambiguous sentences containing a complex NP (e.g., "the son of the doctor") followed by a RC. We constructed 24 sets in a 2x2 design manipulating Main Verb Type and RC position (Tbl. 1). Item varied based on whether the RC modified a complex NP in subject or object position, and whether the main clause contained a *perceptual* (PR-licensing) or *non-perceptual verb*. As PRs are licensed only *following* perceptual verbs, we do not expect Verb Type effects in the Subject RC condition. 94 filler items acted both as attention checks. Since work on Spanish reports faster RTs for perceptual verbs (De Pedis et al. 2025), the experiment comprised a moving window self-paced reading task, followed by a forced-choice comprehension question probing disambiguating preferences (e.g. "*Who run?*").

Experiment 1: Results. Self-identified Italian L1 speakers residing in Italy (n = 120) were recruited through Prolific and completed the study on PCIbex. Items were distributed across participants in a Latin Square design. Each participant's accuracy on the fillers was above 80%. In line with Hemforth et al. (2015), mixed-effects regression analyses reveal a significant effect of RC-position on attachment, with higher HA rates in the Object condition, and the predicted LA preference with RCs in subject position. As expected, we find an interaction effect such that Verb Type affects attachment in the Object condition only. This effect is weaker than what reported by Grillo & Costa (2014): HA rates are lower with non-perceptual verbs, but do not flip the overall trend to LA. Regression analyses on log-transformed RTs at the main verb revealed no effect of Verb Type (Fig. 1,3).

Experiment 2. Lack of a strong(er) LA trend in the Object/Non-Perceptual condition (cf. Grillo & Costa 2014) could be due to SPR encouraging strategies in favor of HA. Thus, we replicated Exp. 1 but without the SPR portion of the task. Participants (n = 120) saw full sentences, and when ready proceeded to a screen with the comprehension question. Results again show a significant effect of RC-position, and an effect of Verb Type in object position. As in Exp. 1, Object/Non-Perceptual items show lower HA rates (Fig. 2). Although we again find no reversal of the overall trend, these results support theories of Perceptual verbs providing cues in favor of HA (and PRs).

Conclusion. The strong LA rates found with RCs in subject position, together with the interaction with Verb Type effects, contribute to problematizing a view of languages as simply "HA" or "LA" (Cotter & Ferreira 2024). Future work should look carefully at the relation between RC position, prosody, and attachment. These results call attention to the need to further explore the multiplicity of factors influencing attachment resolution, in order to reach a fuller understanding of the cognitive mechanisms underlying disambiguation strategies in sentence comprehension.

Table 1: Italian stimuli by condition.

	Verb (PR Avail.)	RC Pos.	Sentence	Predicted
a.	Perceptual (PR/RC)	OBJ	*Gianni vide il figlio del medico che correva la maratona velocemente.* *Gianni saw the son of the doctor who was quickly running the marathon.*	HA > LA
b.	Non-perceptual (RC)	OBJ	*Gianni chiamó il figlio del medico che correva la maratona velocemente.* *Gianni called the son of the doctor who was quickly running the marathon.*	LA > HA
c.	Perceptual (RC)	SBJ	*Il figlio del medico che correva la maratona velocemente vide Gianni.* *The son of the doctor who was quickly running the marathon saw Gianni.*	LA > HA
d.	Non-perceptual (RC)	SBJ	*Il figlio del medico che correva la maratona velocemente chiamó Gianni.* *The son of the doctor who was quickly running the marathon called Gianni.*	LA > HA

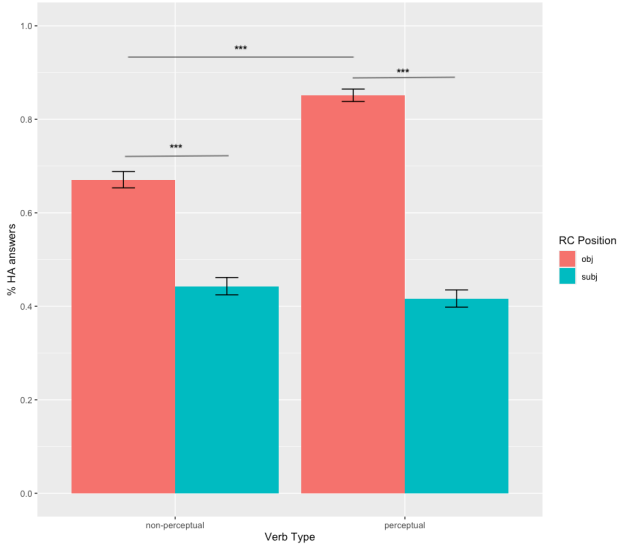

Figure 1: Exp 1. HA answer rate.

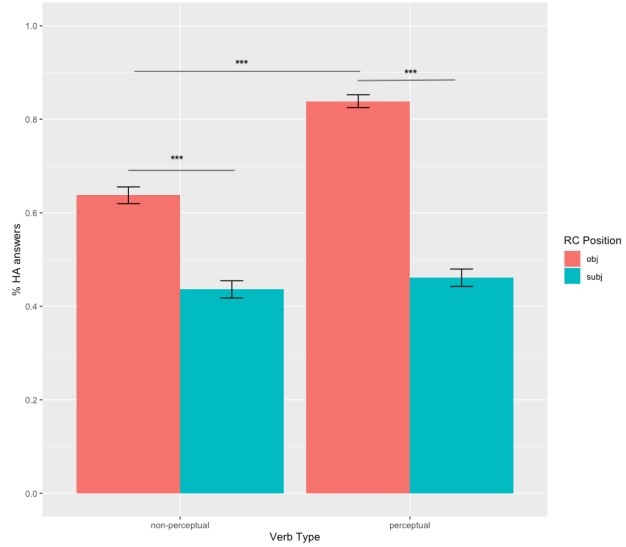

Figure 2: Exp 2. HA answer rate.

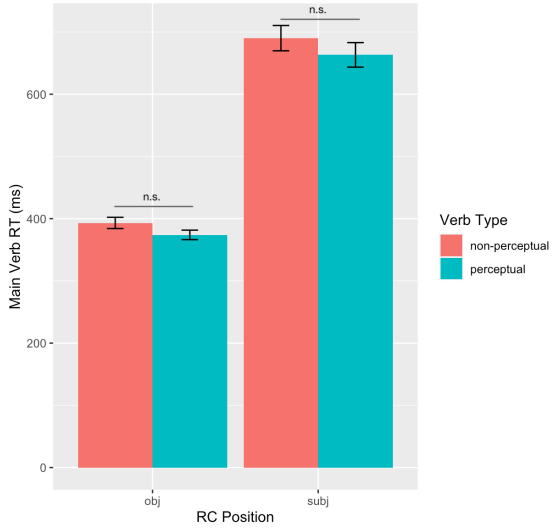

Figure 3: Exp 1. RTs at Matrix Clause Verb.

(1) John saw the son$_{HA}$ of the doctor $_{LA}$ **who was on the balcony**.

Cotter & Ferreira. 2024. Memory & Cognition. **Grillo & Costa**. 2014. Cognition. **De Pedis et al**. 2025. Front. in Psychology. **Hemforth et al**. 2015. Lingua. **Lee & De Santo**. 2024. Language, Cognition, and Neuroscience.